HEICO

Hydraulic & Engineering Instruments

Progressive in Design
Perfect in Performance

OPERATING MANUAL

ANAB ACCREDITED NABCB QM 008

An ISO 9001 : 2015 Company

HEICO

OPERATING INSTRUCTIONS

FOR

U-BOX TEST FINAL

Hydraulic & Engineering Instruments

(An ISO 9001:2000 Company)

B59/4, Naraina Industrial Area, Phase-II,
New Delhi-110 028 (INDIA)

Tel : 91-11-25893820 (4 Lines)

Fax : 91-11-25893152

E-mail : info@heicoin.com

Web : www.heicoin.com

ISO 9001:2000

U Box Test Method for Self Compacting Concrete

U Box test is used to measure the filing ability of self compacting concrete. The apparatus consists of a vessel that is divided by a middle wall into two compartments; an opening with a sliding gate is fitted between the two sections.

Reinforcing bar with nominal diameter of 134 mm are installed at the gate with centre to centre spacing of 50 mm. this create a clear spacing of 35 mm between bars.

The left hand section is filled with about 20 liter of concrete then the gate is lifted and the concrete flows upwards into the other section. The height of the concrete in both sections is measured.

Assessment of U Box test

This is a simple test to conduct, It provides a good direct assessment of filling ability. This is literally what the concrete has to do, modified by an unmeasured requirement for passing ability.

Procedure for U Box Test on Self Compacting Concrete

About 20 liter of concrete is needed to perform the test, sampled normally. Set the apparatus level on firm ground, ensure that the sliding gate can open freely and then close it.

Moisten the inside surface of the apparatus, remove any surplus water, fill the vertical section of the apparatus with the concrete sample.

Leave it stand for 1 minute. Lift the sliding gate and allow the concrete to flow out into the other compartment.

After the concrete has come to rest, measure the height of the concrete in the compartment that has been filled, in two places and calculate the mean (H1).

Measure also the height in the other equipment (H2). Calculate H1-H2, the filling height. The whole test has to be performed within 5 minutes.

Interpretation of the result

If the concrete flows as freely as water, at rest it will be horizontal, so H1-H2=0. Therefore the nearest this test value, the 'filling height', is to zero, the better the flow and passing ability of the concrete.